

Video Transcript

Video 1: Introduction (04:30)

In this section, you have learned a lot about databases, all of the varieties of them. You've learned how to run them in containers. You've looked at multiple languages and how to run within those containers on those different platforms. You've looked at change data capture. So, now it's the time to apply this knowledge and we're going to ask you to do it on transit data from the city of Boston. This is a fun things to do. You're pulling data directly from the source. This is real data of real systems moving out in the world. And so, you will be making the decision of what database do I use? How do I handle data capture? How do I update the front end?

And in this case, we'll start you off with a stock one, but you can take all of this as always up to your own satisfaction and as far as you would like. The internet of things, IoT is the third generation of the web. The first one was about connecting documents. The second one was extending this connectivity to programs and data. And the third one is reaching out onto the physical world, the physical internet. And so, you can see here that a broad category of this third generation of the web is location-based applications. It turns out that there's a great deal of intelligence that can be derived and also great services that can be offered.

Now more specifically, we're going to be looking at bus tracking. And that is one of those things that if in areas like the Boston area on a cold winter night, especially if you have a storm, as you see here in the picture, you do not want to sit out in the cold. And now to be fair, also in places where it's very hot, like let's say the California desert, the area of death valley, the same thing would be true.

Now we're going to be taking a look at one specific route. This is Boston MBTA Route 1, MBTA stands for Massachusetts Bay Transportation Authority. And, this is a bus line that runs, it's a little bit more than that, but for practical purposes, we will define it as running between three university campus. And you can see there that it's between Harvard, MIT, and Boston University. And so once again, what we want to do is we want to be able to know where these buses are for the convenience of users. We don't want people to freeze. And as mentioned as well, we want to capture some of that data. It would provide us to do analysis. It would allow those that are

decision makers to see the performance, they would be able to see a historical record. And there are a number of advantages of doing that.

And so, the project that we're going to do is going to have the following architecture. You're going to have a web application that connects to a server that you will be writing. Now that server will be connecting to MBTA, and MBTA has provided an interface through which you can ask about the location of their buses. So, this is the portal to the developer portal to the services that are provided by MBTA. Now as mentioned, your part will be the server. We will provide you with some basic functionality on a web app that will connect to an API that you will have to write, an application programming interface that will be running on a server that you will be creating.

Now you will also be responsible for interrogating MBTA and requesting that information. So, more formally, your server will need to query MBTA, will need to store that data, will need to refresh that data periodically. It will need to build an API for clients, and it will need to serve that data on demand. Now once you've done the basics, you can feel free to expand onto the web application or expand on what you're doing on the server. There's a lot that can be done there, but we want you to get started first with the basics and then if you have time, you can do more.